

## Data Flow Diagrams (DFD)

### 1. Level 0: Context Diagram

User (Meteorologist/Relief Coordinator) submits Cloud Cover and Rainfall inputs. The Rising Waters system processes inputs and outputs a binary risk classification (High Flood Risk / Low Flood Risk) along with a model confidence percentage.

### 2. Level 1: Process Architecture Flow

- User submits HTML form.
- Flask Server intercepts request.
- Validation Engine tests boundary rules.
- Feature scaling applies StandardScaler parameters.
- Decision Tree Classifier evaluates June-September threshold (2425.64 mm).
- Sigmoid confidence algorithm calculates confidence score.
- AJAX updates HTML with Risk Audit list without reloading page.